Reg. No. | | | | | | | | | | |

[17EE44]

FOURTH SEMESTER B.Tech EXAMINATION, JUNE–2019

POWER GENERATION AND UTILIZATION OF ELECTRICAL POWER

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least two full questions from each part.
2. Missing data may be suitably assumed.

PART-A

1. a. Discuss briefly the classification of hydro-electric plants. **05**

   b. Explain the general arrangements and operation of a hydro electric plant. **05**

   c. State the factors for selection of a site for hydro electric stations. **04**

2. a. Write any two advantages and disadvantages of thermal power plant. **04**

   b. Explain the working of a thermal power plant with the help of typical layout. **10**

3. a. With the help of block diagram, Explain the operation of a Nuclear power Plant. What are its merits and demerits? **10**

   b. What are the factors considered for site selection of a Nuclear Power Plant? **04**

4. a. Describe briefly the main components of a diesel electric plant. **08**

   b. Explain with schematic layout the working of a gas turbine plant. **06**

PART - B

5. a. Define the following terms in connection with a power supply system: **08**

   i. connected load

   ii. Maximum demand

   iii. Demand Factor

   iv. Load factor. Discuss the effect of load factor on the cost of generation in a power system.

   b. A generating station supplies the following loads to various consumers: **06**

   Industrial consumer = 750MW;

   Commercial establishment = 350MW;

   Domestic power=10MW;

   Domestic light=50MW;

   If the maximum demand on the station is 1000 MW and the number of kwh generated per year is 50×10^5 , determine i) the diversity factor and ii) annual load factor

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

6. a. Discuss the principle of arc welding. Differentiate between carbon and metallic arc welding. **08**

b. Determine the efficiency of a high frequency induction furnace which takes 10 minutes to melt 1.815 Kg of aluminium, the input to the furnace being 5 KW and the initial temperature 15^0 C. **06**

Specific heat of aluminium=0.212

Melting point=660^0C

Latent heat of fusion of aluminum=76.8 K cal/Kg.

7. a. What are the various types of starters used for fluorescent tubes? Describe their construction and principle of operation. **08**

b. A building measuring 30m×20m is to be floodlit on the front side with brightness of 25 lumen/sq.metre. Coefficient of reflection of building surface is 0.25. Lamps of 500 W having lumens output of 8000 each are used. Assuming beam factor as 0.6, waste light factor 1.2 and maintenance factor as 0.75, determine the number of lamps required. **06**

8. a. Explain the necessity and applications of refrigeration systems. **08**

b. Write a short note on eco-friendly refrigerants. **06**